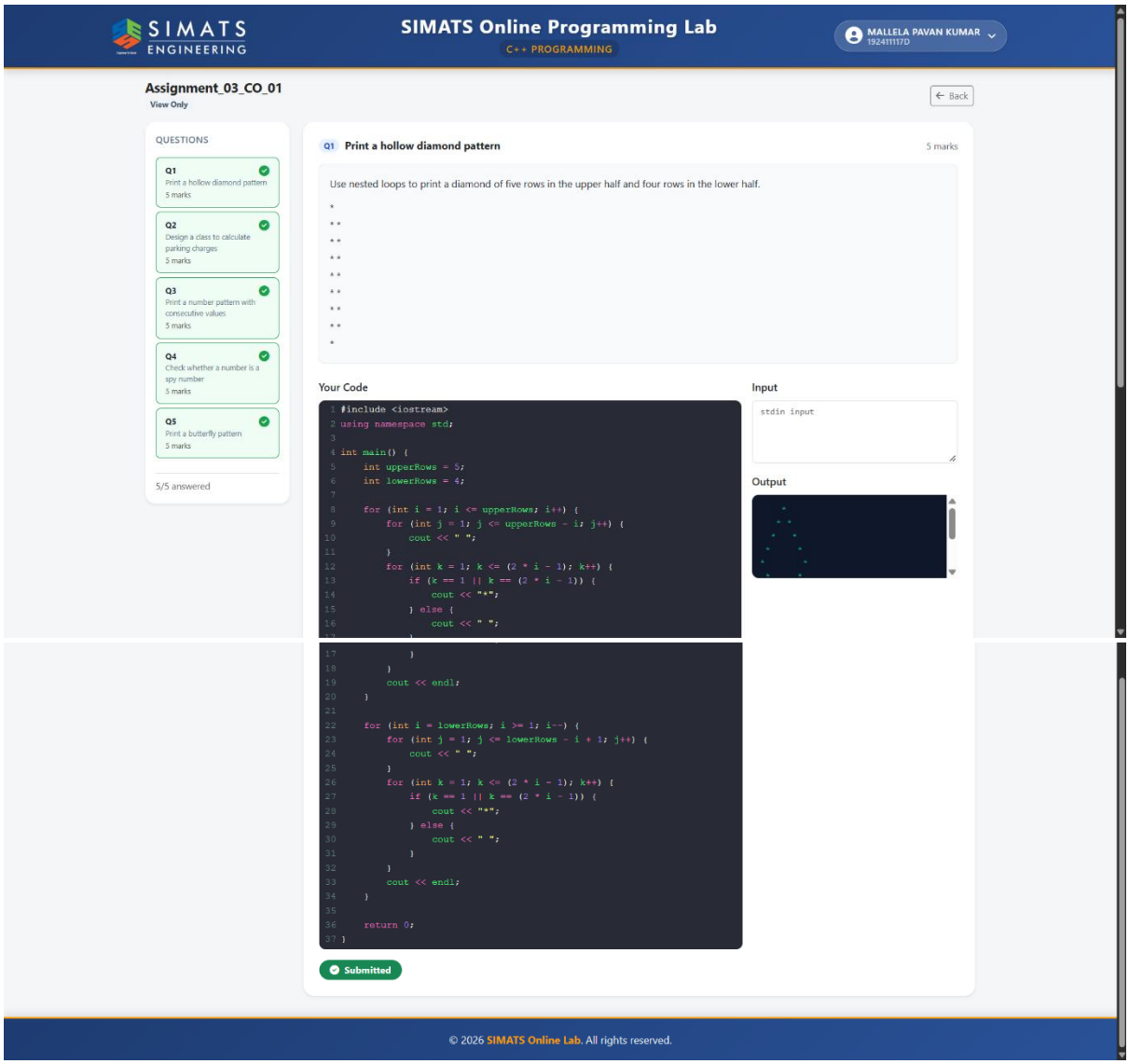
M.Pavan kumar 192411117

DSA0120 SLOT-D

# Assignment_03_CO_01

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SIMATS ENGINEERING

# SIMATS Online Programming Lab

C++ PROGRAMMING

MALLELA PAVAN KUMAR
192411117D

## Assignment_03_CO_01

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QUESTIONS

- Q1 Print a hollow diamond pattern — 5 marks
- Q2 Design a class to calculate parking charges — 5 marks
- Q3 Print a number pattern with consecutive values — 5 marks
- Q4 Check whether a number is a spy number — 5 marks
- Q5 Print a butterfly pattern — 5 marks

5/5 answered

### Q2 Design a class to calculate parking charges

5 marks

Create a Parking class. Charge ₹30 for the first hour and ₹20 for every additional hour. Any fraction of an hour counts as a full hour.
Example: 2 hours and 15 minutes → ₹70.
TC_01:2 15->Parking charge: Rs. 70.TC_02:0 30->Parking charge: Rs. 30.TC_03:2 60->Invalid duration

Your Code

```cpp
#include <iostream>
#include <cmath>

using namespace std;

class Parking {
public:
    void calculateCharge(int hours, int minutes) {

        if (hours < 0 || minutes < 0 || minutes >= 60) {
            cout << "Invalid duration" << endl;
            return;
        }

        if (hours == 0 && minutes == 0) {
            cout << "Parking charge: Rs. 0" << endl;
            return;
        }

        int totalHours = hours;
        if (minutes > 0) {
            totalHours += 1;
        }

        // Base rate logic
        int charge = 0;
        if (totalHours >= 1) {
            charge += 30;
        }
        if (totalHours > 1) {
            charge += (totalHours - 1) * 20;
        }

        cout << "Parking charge: Rs. " << charge << endl;
    }
};

int main() {
    Parking tracker;

    cout << "TC_01: ";
    tracker.calculateCharge(2, 15);

    cout << "TC_02: ";
    tracker.calculateCharge(0, 30);

    cout << "TC_03: ";
    tracker.calculateCharge(2, 60);

    return 0;
}
```

Input

```
stdin input
```

Output

```
TC_01: Parking charge: Rs. 70
TC_02: Parking charge: Rs. 30
TC_03: Invalid duration
```

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

0.00 / 5 marks

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- Q5 Print a butterfly pattern — 5 marks

5/5 answered

### Q3 Print a number pattern with consecutive values

5 marks

Print the following pattern for four rows.

1
2 3
4 5 6
7 8 9 10

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 1;

    for (int i = 1; i <= 4; i++) {
        for (int j = 1; j <= i; j++) {
            cout << n;
            if (j < i) cout << " ";
            n++;
        }
        cout << endl;
    }

    return 0;
}
```

Input

```
stdin input
```

Output

```
1
2 3
4 5 6
7 8 9 10
```

Submitted

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- **Q5** Print a butterfly pattern — 5 marks

5/5 answered

### Q4 Check whether a number is a spy number

5 marks

A number is a spy number if the sum of its digits equals the product of its digits.
Example: 1124 → Sum = 8, product = 8.
TC_01:1124->Spy number.TC_02:124->Not a spy number.TC_03:-123->Invalid input

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    long long num;

    if (!(cin >> num)) {
        cout << "Invalid input" << endl;
        return 0;
    }

    if (num < 0) {
        cout << "Invalid input" << endl;
        return 0;
    }


    if (num == 0) {
        cout << "Spy number" << endl;
        return 0;
    }

    long long temp = num;
    long long sum = 0;
    long long product = 1;


    while (temp > 0) {
        long long digit = temp % 10;
        sum += digit;
        product *= digit;
        temp /= 10;
    }


    if (sum == product) {
        cout << "Spy number" << endl;
    } else {
        cout << "Not a spy number" << endl;
    }

    return 0;
}
```

Input

```
1124
```

Output

```
Spy number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

Submitted

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- **Q5** Print a butterfly pattern — 5 marks

5/5 answered

### Q5 Print a butterfly pattern — 5 marks

Print a butterfly pattern using nested loops.

```
* *
** **
*** ***
**** ****
**********
**** ****
*** ***
** **
* *
```

**Your Code**

```cpp
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 5; i++) {
        for (int j = 1; j <= i; j++)
            cout << "*";
        for (int j = 1; j <= 2 * (5 - i); j++)
            cout << " ";
        for (int j = 1; j <= i; j++)
            cout << "*";
        cout << endl;
    }

    for (int i = 4; i >= 1; i--) {
        for (int j = 1; j <= i; j++)
            cout << "*";
        for (int j = 1; j <= 2 * (5 - i); j++)
            cout << " ";
        for (int j = 1; j <= i; j++)
            cout << "*";
        cout << endl;
    }

    return 0;
}
```

**Input**

stdin input

**Output**

```
*         *
**       **
***     ***
****   ****
**********
****   ****
```

Submitted

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